Recitation Assignment #4

October 16, 2024

You may complete this in class. However, if you are unable to do so, it is expected that you complete this for recitation next time.

If you have any questions, please ask.

Today, we are going to model masses on springs, and model simple harmonic motion. I will give you considerably less guidance on writing this program than the previous time, so you should definitely have a copy of the VPython website open, and last week's recitation assignment in front of you.

New Concepts:

Physical:

- Oscillators

Computational:

- New VPython Object: Helix
- User Input
- Timing

1. Let's begin by creating a new program/Jupyter notebook, with the following commands at the top (after "from vpython import *")

```
L=3 #The unstretched length of the spring

yceiling=1.5 #The position of the ceiling

g=9.8

y0=yceiling-L

ceiling=box(pos=vec(0,yceiling,0),length=6,width=4,height=0.2,texture=textures.granite)

scene.autoscale=0

spring1=helix(pos=vec(0,yceiling,0),axis=vec(0,-L,0),radius=0.1*L)

ball1=sphere(pos=vec(0,y0,0),radius=0.2*L)
```

This simply sets a number of parameters and draws a mass on a spring. You need not include the comments.

This program sets a number of fairly self-explanatory parameters and draws a mass hanging from a spring. The spring is drawn using the VPython "helix" object. Make sure you read the documentation on it.

2. For this assignment, you will need to read in user input for:

    - The spring constant
    - The mass
    - The initial displacement of the oscillator

For instance, to read in a spring constant, you'll use:

```
K=float(input("What is the spring constant (N/m)?"))
```

Where the "float" around everything tells python to read what you type as a number.

Allow the user to enter in all 3 parameters.

3. Now, you'll need to include the physics. Use the example from last week to create a "while" loop. Run the code for 20 seconds with timesteps of 0.01 seconds.

At each timestep, you'll want to compute the force on the block with a command like:

```
ball1.F=...
```

Note that the force calculation is indented, because you're under the "while" loop. Remember, there are two forces on the block, gravity, and the spring:

$$\vec{F} = -mg\hat{\jmath} - k(y - y_0)\hat{\jmath}$$

Where y_0 is the equilibrium position of the spring that we defined before. In reality, those terms will look something like:

```
(ball.pos-vec(0,y0,0))
```

Do not simply put this in verbatim! After all, it's not an equation. The later examples are going to be more complicated, which means you need to understand what you are doing.

4. Update the velocity of the ball (you did this last week as well – be sure to glance at your work from previous weeks to be reminded of past work):

$$\vec{v} = \vec{v} + \frac{\vec{F}}{m}\Delta t$$

Make sure you label the terms correctly.

5. Update the velocity of the ball

$$\vec{r} = \vec{r} + \vec{v}\Delta t$$

6. Update the position of the bottom of the spring. You'll want to adjust "spring1.axis" appropriately. This doesn't include any additional physics. It just makes the simulation look nice.

7. This is the hard part. Using a "flag," you'll want to tell the program to output the time when the ball has undergone one full oscillation. Here's my suggested algorithm:
    a. If the flag is set to 0 (which it is initially) AND the ball is moving upwards, set the flag to 1.
    b. If the flag is set to 1 (which it will be only after it's started moving upwards), and the ball is moving down, set the flag to 2, and print out, or record in a variable, the current time. That should be the period of half of a full cycle. Remember that a full cycle is when the oscillator has reached its original position again.
8. Use the relationships we developed in class:

$$\omega^2 = \frac{k}{m}$$

and

$$T = \frac{2\pi}{\omega}$$

to compute the theoretical period of oscillations. Print this value out and compare to the "measured" value of the oscillations. Is your estimate good enough?

Run your code with 3 combinations of k and m. Include both the "measured" period along with the theoretical one in a markdown cell within your notebook (or in the text of your email to Kendall, if you're not using Jupyter).

As always, send your code to Kendall and Professor Cruz via email.